Abu Dhabi City Campus Daily/Weekly Planner

Week 9: 26th – 30th October 2026

Subject: Mathematics **Unit Title: Differentiation** **Grade: 12**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Estimating Derivatives of a Function at a Point	I can estimate the derivative of a function at a point using a graphing calculator, a tangent-line slope, or data presented in a table.	2.1 Clarify derivative vocabulary and notation using graphs, tables, and worked examples. 2.3 Support decoding of derivative notation and calculator inputs through step-by-step modelling.	Related Videos / Practice 2.3 Estimating Derivatives of a Function at a Point.mp4 2.4 Connecting Differentiability and Continuity.mp4
Lesson 2: Connecting Differentiability and Continuity	I can explain the relationship between continuity and differentiability and identify points where a derivative does not exist.	2.1 Clarify differentiability and continuity using graphs and key vocabulary. 2.3 Connect graphical features to whether a derivative exists.	
Lesson 3: Review – Limits & Continuity	I can apply limits and continuity concepts to solve mixed problems involving the Squeeze Theorem, Intermediate Value Theorem, discontinuities, and continuity.	2.3 Support connections between graphs, tables, and algebraic representations. 5.3 Build fluency through graduated mixed review and worked examples.	
Lesson 4: Summative Assessment – Limits & Continuity	I can independently demonstrate and justify my understanding of limits and continuity using graphical, numerical, and analytical representations.	5.3 Build fluency through independent assessment and careful checking. 8.2 Use a range of question types to assess foundational and higher-order understanding.	
Homework	Revision sheets and practice resources posted on Teams/OneNote.		
Teacher's notes/ Reminders	Microsoft Teams and OneNote; lesson PowerPoints; review sheets; GDC/Desmos; AP Calculus practice and assessment materials.		

Technology Integration	Microsoft Teams; OneNote; GDC/graphing calculator (especially numerical derivative function); Desmos; lesson PowerPoints; digital practice and assessment materials.
Cross-curricular links	**Physics/Science:** interpret derivatives as rates of change in motion and other changing quantities.
External Assessment Integration	Direct alignment with AP Calculus AB

Wellbeing/ ATL Skills	**Thinking Skills:** Interpret graphs and tables, estimate derivatives, connect differentiability with continuity, and justify conclusions. **Communication Skills:** Use precise mathematical vocabulary and notation to explain whether a derivative exists and whether a function is continuous. **Self-Management Skills:** Organise revision, manage assessment time, check work, and reflect on misconceptions.
Standards	AP Calculus AB: 2.3 Estimating Derivatives of a Function at a Point; 2.4 Connecting Differentiability and Continuity
Inquiry/ TOK Question	How do graphical and numerical evidence help us estimate change and decide whether a function is continuous or differentiable?
Success Criteria (Measurable)	**Lesson 1: Estimating Derivatives of a Function at a Point** • **All students must** estimate a derivative at a point from a graph or table and identify it as the slope of the tangent line. • **Most students should** estimate derivatives from tables and use a GDC to calculate a numerical derivative at a specified x-value. • **Some students could** compare estimates from different representations and interpret the derivative in a real-world context. **Lesson 2: Connecting Differentiability and Continuity** • **All students must** identify points where a function is continuous or discontinuous and where a derivative does not exist. • **Most students should** explain why discontinuities, corners/cusps, and vertical tangents prevent differentiability at a point. • **Some students could** justify the relationship that differentiability implies continuity and explain why continuity does not guarantee differentiability. **Lesson 3: Review – Limits & Continuity** • **All students must** evaluate limits and identify continuity/discontinuity from graphs, tables, or equations, and state the key conditions of the Squeeze Theorem and IVT. • **Most students should** apply the Squeeze Theorem and IVT with appropriate conditions and justify conclusions about continuity and discontinuities. • **Some students could** solve unfamiliar mixed problems involving piecewise functions, parameters, and asymptotic behaviour, with complete mathematical justification. **Lesson 4: Summative Assessment – Limits & Continuity** • **All students must** demonstrate foundational skills in evaluating limits and identifying continuity, discontinuities, and asymptotes. • **Most students should** apply the Squeeze Theorem and IVT and justify conclusions using correct notation and conditions. • **Some students could** solve multi-step or unfamiliar problems integrating limits, continuity, discontinuities, and asymptotic behaviour.
National Identity Domain	Respect الاحترام Area: Citizenship Implementation: Encourage mathematical integrity by showing working, using accurate notation, and supporting conclusions with evidence during review and assessment.
	Area: Citizenship

	Implementation: Promote responsibility for learning by using feedback and error analysis to address misconceptions and by completing the summative assessment independently and honestly.